

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

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भारतीय मानक मसौदा
**चिकित्सीय उपयोग के लिए सुई-रहित इंजेक्शन प्रणालियाँ — अपेक्षाएँ
और परीक्षण पद्धतियाँ**

Draft Indian Standard
**Needle-Free Injection Systems for Medical Use — Requirements
and Test Methods**

[ICS 11.040.20]

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

Last date for comments:
19 September 2026

NATIONAL FOREWORD

(Adoption clause will be added later)

The text of ISO Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are however not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- Wherever the words ‘International Standard’ appear referring to this standard, they should be read as ‘Indian Standard’; and
- Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are listed below along with their degree of equivalence for the editions indicated:

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
ISO 7886-3 : 2020 Sterile hypodermic syringes for single use — Part 3: Auto-disabled syringes for fixed-dose immunization	IS 10258 (Part 3) : 2021/ISO 7886-3 : 2020 Sterile hypodermic syringes for single use: Part 3 Auto-disabled syringes for fixed-dose immunization (<i>first revision</i>)	Identical
ISO 10993-4 Biological evaluation of medical devices	IS/ISO 10993-4 : 2017 Biological evaluation of medical	Identical

— Part 4: Selection of tests for interactions with blood	devices: Part 4 Selection of tests for interactions with blood	
ISO 10993-5 Biological evaluation of medical devices — Part 5: Tests for in vitro cytotoxicity	IS/ISO 10993-5 : 2009 Biological evaluation of medical devices: Part 5 Tests for in vitro cytotoxicity	Identical
ISO 10993-9 Biological evaluation of medical devices — Part 9: Framework for identification and quantification of potential degradation products	IS 17932 (Part 2) : 2024 Biological evaluation of medical devices: Part 2 Framework for identification and quantification of potential degradation products (ISO 10993-9 : 2019, MOD)	Modified
ISO 10993-10 Biological evaluation of medical devices — Part 10: Tests for skin sensitization	IS 17932 (Part 6) : 2023 Biological evaluation of medical devices: Part 6 Tests for skin sensitization (ISO 10993-10 : 2021, MOD)	Modified
ISO 10993-11 Biological evaluation of medical devices — Part 11: Tests for systemic toxicity	IS/ISO 10993-11 : 2017 Biological evaluation of medical devices: Part 11 Tests for systemic toxicity	Identical
ISO 10993-12 Biological evaluation of medical devices — Part 12: Sample preparation and reference materials	IS/ISO 10993-12 : 2021 Biological evaluation of medical devices: Part 12 Sample preparation and reference materials	Identical
ISO 10993-13 Biological evaluation of medical devices — Part 13: Identification and quantification of degradation products from polymeric medical devices	IS/ISO 10993-13 : 2010 Biological evaluation of medical devices: Part 13 Identification and quantification of degradation products from polymeric medical devices	Identical
ISO 10993-14 Biological evaluation of medical devices — Part 14: Identification and quantification of degradation products from ceramics	IS/ISO 10993-14 : 2001 Biological evaluation of medical devices: Part 14 Identification and quantification of degradation products from ceramics	Identical
ISO 10993-15 Biological evaluation of medical devices — Part 15: Identification and quantification of degradation	IS 17932 (Part 3) : 2024 Biological evaluation of medical devices: Part 3 Identification and quantification of degradation	Modified

products from metals and alloys	products from metals and alloys (ISO 10993-15 : 2019, MOD)	
ISO 10993-16 Biological evaluation of medical devices — Part 16: Toxicokinetic study design for degradation products and leachables	IS/ISO 10993-16 : 2017 Biological evaluation of medical devices: Part 16 Toxicokinetic study design for degradation products and leachables	Identical
ISO 10993-18 Biological evaluation of medical devices — Part 18: Chemical characterization of medical device materials within a risk management process	IS/ISO 10993-18 : 2020 Biological evaluation of medical devices: Part 18 Chemical characterization of medical device materials within a risk management process	Identical
ISO/TS 10993-19 Biological evaluation of medical devices — Part 19: Physico-chemical, morphological and topographical characterization of materials	IS 17932 (Part 4) : 2024 Biological evaluation of medical devices: Part 4 Physico-chemical, morphological and topographical characterization of materials (ISO 10993-19 : 2020, MOD)	Modified
ISO/TS 10993-20 Biological evaluation of medical devices — Part 20: Principles and methods for immunotoxicology testing of medical devices	IS/ISO/TS 10993-20 : 2006 Biological evaluation of medical devices: Part 20 Principles and methods for immunotoxicology testing of medical devices	Identical
ISO/TR 10993-22 Biological evaluation of medical devices — Part 22: Guidance on nanomaterials	IS/ISO/TR 10993-22 : 2017 Biological evaluation of medical devices: Part 22 Guidance on nanomaterials	Identical
ISO 10993-23 Biological evaluation of medical devices — Part 23: Tests for irritation	IS 17932 (Part 7) : 2024 Biological Evaluation of Medical Devices: Part 7 Tests for irritation (ISO 10993-23 : 2021, MOD)	Modified
ISO 11201 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions in an essentially free field over a	IS 19549 : 2026/ISO 11201 : 2010 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions in an essentially free field over a	Identical

reflecting plane with negligible environmental corrections	reflecting plane with negligible environmental corrections	
ISO 11202 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions applying approximate environmental corrections	IS 19551 : 2026/ISO 11202 : 2010 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions applying approximate environmental corrections	Identical
ISO 11204 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions applying accurate environmental corrections	IS 19547 : 2026/ISO 11204 : 2010 Acoustics — Noise emitted by machinery and equipment — Determination of emission sound pressure levels at a work station and at other specified positions applying accurate environmental corrections	Identical
ISO 14971 : 2019 Medical devices — Application of risk management to medical devices	IS/ISO 14971 : 2019 Medical devices — Application of risk management to medical devices (<i>first revision</i>)	Identical
IEC 60068-2-27 Environmental testing — Part 2: Tests. Test Ea and guidance: Shock	IS 9000 (Part 7/Sec 1) : 2018/IEC 60068-2-27 : 2008 Basic environmental testing procedures for electronic and electrical items: Part 7 Impact test Section 1 Shock (Test Ea) (<i>second revision</i>)	Identical
IEC 60068-2-31 Environmental testing — Part 2-31: Tests. Test Ec: Rough handling shocks, primarily for equipment-type specimens	IS 9000 (Part 7/Sec 3) : 2019/IEC 60068-2-31 : 2008 Environmental testing: Part 7 Tests Section 3 Test Ec: Rough handling shocks, primarily for equipment-types specimens (<i>first revision</i>)	Identical
IEC 60529 Degrees of protection provided by enclosures (IP Code)	IS/IEC 60529 : 2001 Degrees of protection provided by enclosures (IP Code)	Identical

IEC 60721-3-7 : 1995+AMD1 : 1996 Classification of environmental conditions — Part 3-7: Classification of groups of environmental parameters and their severities — Portable and non-stationary use	IS/IEC 60721-3-7 : 2002 Classification of environmental conditions: Part 3 Classification of groups of environmental parameters and their severities: Section 7 Portable and non-stationary use	Identical
IEC 61000-4-2 : 2008 Electromagnetic compatibility (EMC) — Part 4-2: Testing and measurement techniques — Electrostatic discharge immunity test	IS 14700 (Part 4/Sec 2) : 2018/IEC 61000-4-2 : 2008 Electromagnetic compatibility (EMC): Part 4 Testing and measurement techniques: Section 2 Electrostatic discharge immunity test (<i>second revision</i>)	Identical
IEC 61000-4-3 : 2020 Electromagnetic compatibility (EMC) — Part 4-3: Testing and measurement techniques — Radiated, radio-frequency, electromagnetic field immunity test	IS 14700 (Part 4/Sec 3) : 2023/IEC 61000-4-3 : 2020 Electromagnetic compatibility (EMC): Part 4 Testing and measurement techniques: Section 3 Radiated, radio-frequency electromagnetic field immunity test (<i>second revision</i>)	Identical
IEC 61672-1 Electroacoustics — Sound level meters — Part 1: Specifications	IS 15575 (Part 1) : 2016/IEC 61672-1 : 2013 Electroacoustics — Sound level meters: Part 1 Specifications (<i>first revision</i>)	Identical
IEC 62366-1 Medical devices — Part 1: Application of usability engineering to medical devices	IS 17922 (Part 1) : 2023/IEC 62366-1 : 2015 + AMD 1 : 2020 Medical devices: Part 1 Application of usability engineering (<i>first revision</i>)	Identical

The Committee responsible for the preparation of this standard has reviewed the provisions of the following International Standards referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

<i>International Standard/ Other Publication</i>	<i>Title</i>
ISO 10993-1	Biological evaluation of medical devices — Part 1: Requirements and general principles for the evaluation of biological safety within a risk management process

ISO 10993-2	Biological evaluation of medical devices — Part 2: Animal welfare requirements
ISO 10993-3	Biological evaluation of medical devices — Part 3: Evaluation of genotoxicity, carcinogenicity, reproductive toxicity and developmental toxicity
ISO 10993-6	Biological evaluation of medical devices — Part 6: Tests for local effects after implantation
ISO 10993-7	Biological evaluation of medical devices — Part 7: Ethylene oxide sterilization residuals
ISO 10993-17	Biological evaluation of medical devices — Part 17: Toxicological risk assessment of medical device constituents
ISO/TR 10993-55	Biological evaluation of medical devices — Part 55: Interlaboratory study on cytotoxicity
ISO 14155	Clinical investigation of medical devices for human subjects — Good clinical practice
IEC 60068-2-64	Environmental testing — Part 2-64: Tests — Test Fh: Vibration, broad-band random and guidance

This standard also makes a reference to the BIS Certification Marking of the product. Details of which is given in National Annex A.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

NATIONAL ANNEX A (National Foreword)

A-1 BIS CERTIFICATION MARKING

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

Scope

This document applies to safety and performance and testing requirements for single-use and multiple-use Needle-Free Injection Systems (NFISs) intended for human use in clinics and other medical settings and for personal use by patients.

The dose chamber of the NFIS is often disposable and intended to be replaced after either a single use or a limited number of uses. It is sometimes separable from the injection mechanism and often termed a “cartridge”, “ampoule”, “syringe”, “capsule” or “disc”. In contrast, the dose chamber can also incorporate a permanent internal chamber designed to last through the claimed life of the device, and an additional member or members which eliminate the risk of cross-contamination.

Excluded from this document are drug delivery methods which:

- involve penetration of a part of the device itself into or through skin or mucous membranes (such as needles, tines, micro-needles, implantable slow-release drug devices);
- generate aerosols, droplets, powders or other formulations for inhalation, insufflation, intranasal or oral deposition (such as sprays, inhalers, misters);
- deposit liquids, powders, or other substances on the surface of skin or mucosal surfaces for passive diffusion or ingestion into the body (such as transdermal patches, liquid drops);
- apply sonic or electromagnetic energy (such as ultrasonic or iontophoretic devices);
- infusion systems for adding or metering medication into or through systems of artificial tubes, catheters, and/or needles which themselves enter the body.

The technical content of the document has not been enclosed as it is identical with the corresponding ISO standard. For details, please refer to ISO 21649 : 2023 or kindly contact:

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